

9.13 Advancements in NC machines

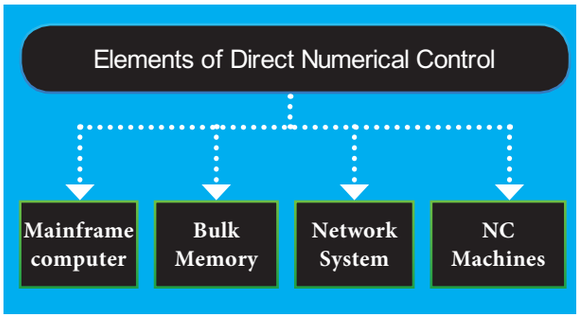
After the modern micro-processor based computers took control of the NC system, the technology in this field has grown tremendously. Following are two major advancements of NC system.

1. Direct Numerical control(DNC)
2. Adaptive control(AC)

Direct Numerical control (DNC)

A manufacturing system in which several NC machines are connected to and controlled by a remotely located main frame computer is known as 'Direct Numerical Control' (DNC). The basic constituents are

a. Main frame computer
b. Bulk memory
c. Network system
d. NC machine tools



DNC system eliminates the need of separate machine control units, input media, this system governs the scheduling of work of all the NC machine tools. The required network system may be a LAN (Local Area Network) or WAN (Wide Area Network)

Adaptive control

During machining, cutting speed, feed and depth of cut are measured and compared with desired values.

If it is not so, it will adapted (modified), then it is called as "Adaptive control".

ACTIVITIES

1. Students to visit the polytechnic, Engineering colleges for showing the CNC machines and its functioning.

Questions

Part I.

Choose the correct option 1 Mark

1. Production is achieved by
   a) man and machine
   b) materials and tools
   c) men, machine, materials and tools
   d)men and tools

2. In NC system the program instructions are given as,
   a) Information
   b)numerical data
   c) symbols
   d)thread symbols

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